King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 208 Major Exam II Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. An appropriate form of a particular solution y_p for the non-homogeneous differential equation $y^{(3)} - 8y = e^{2x} + 3x$ is given by $y_p(x) =$

(a) $Axe^{2x} + Bx + C$

(b) $Ax^2e^{2x} + Bx + C$

(c) $Ae^{2x} + Bx^2 + C$

(d) $Ae^{2x} + Bx + C$

(e) $Ae^{2x} + Bx^2 + Cx$

2. The rank of the matrix $A = \begin{bmatrix} 1 & 3 & 3 & 9 \\ 2 & 7 & 4 & 8 \\ 2 & 7 & 5 & 12 \\ 2 & 8 & 3 & 2 \end{bmatrix}$ is

(a) 1

(b) 2

(c) 4

(d) 3

(e) 0

3. If $y = c_1e^{Ax} + (c_2 + c_3x + c_4x^2)e^{Bx}$ is the general solution of the differential equation

$$y^{(4)} + y^{(3)} - 3y'' - 5y' - 2y = 0,$$

then $A - B =$

(a) 4

(b) -2

(c) 0

(d) 3

(e) 2

4. The wronskian of the functions

$$f(x) = e^{2x},\ g(x) = \cosh x,\ h(x) = \sinh x \text{ on } (-\infty, \infty)$$

is

(a) $3e^{2x}$

(b) $5e^{2x}$

(c) 0

(d) $4e^{2x}$

(e) e^{2x}

5. The characteristic polynomial of the matrix $\begin{bmatrix} 5 & 0 & 0 \\ 4 & -4 & -2 \\ -2 & 12 & 6 \end{bmatrix}$ is $p(\lambda) =$

(a) $\lambda^3 - 7\lambda^2 + 6\lambda$

(b) $\lambda^3 - 6\lambda^2 + 10\lambda$

(c) $-\lambda^3 + 7\lambda^2 - 10\lambda$

(d) $-\lambda^3 - 7\lambda^2 + 9\lambda$

(e) $-\lambda^3 + 7\lambda^2 - 8\lambda$

6. A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = Ae^{3x} + Bxe^{3x} + C\cos(4x) + D\sin(4x)$$

is

(a) $y^{(4)} - 6y''' + 25y'' - 96y' + 144y = 0$

(b) $y^{(4)} - 6y''' - 25y'' + 96y' + 144y = 0$

(c) $y^{(4)} + 6y''' - 25y'' - 96y' + 144y = 0$

(d) $y^{(4)} - 6y''' + 25y'' + 96y' - 144y = 0$

(e) $y^{(4)} + 4y''' - 25y'' + 96y' - 144y = 0$

7. If the solution space of the system

$$\begin{aligned} x_1 - 4x_2 - 3x_3 - 7x_4 &= 0 \\ 2x_1 - x_2 + x_3 + 7x_4 &= 0 \\ x_1 + 2x_2 + 3x_3 + 11x_4 &= 0 \end{aligned}$$

has all linear combinations of the two vectors $u = (-1, -1, \alpha, 0)$ and $v = (\beta, -3, 0, 1)$ then $\alpha - \beta =$

(a) 4

(b) 6

(c) 0

(d) 5

(e) -4

8. Which one of the following sets of functions is linearly dependent?

(a) $y_1(x) = e^x,\ y_2(x) = e^{-x},\ y_3(x) = e^{2x}$

(b) $y_1(x) = 1,\ y_2(x) = x,\ y_3(x) = 1 + x^2$

(c) $y_1(x) = \sin x,\ y_2(x) = \cos x,\ y_3(x) = e^x$

(d) $y_1(x) = x,\ y_2(x) = x^2,\ y_3(x) = x^3$

(e) $y_1(x) = e^x,\ y_2 = e^{-x},\ y_3 = \sinh x$

9. The solution of the initial-value problem

$$y'' + 4y = 0; \quad y(0) = 3 \text{ and } y'(0) = 8$$

is

(a) $y = 3\cos(2x) - 4\sin(2x)$

(b) $y = 3\cos(x) + 4\sin(x)$

(c) $y = 3\cos(4x) + 4\sin(4x)$

(d) $y = 3\cos(2x) + 2\sin(2x)$

(e) $y = 3\cos(2x) + 4\sin(2x)$

10. Consider the vectors

$$v_1 = (-4, 2, -7),\ v_2 = (3, -1, 3) \text{ and } w = (7, 1, -8).$$

If $w = av_1 + bv_2$, then $a + b =$

(a) 10

(b) 8

(c) 12

(d) 14

(e) 6

11. If the matrix $A = \begin{bmatrix} 5 & -4 \\ 3 & -2 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

12. The eigenvector associated with the eigenvalue $\lambda = -i$ of the matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ is $v_1 = \begin{bmatrix} i \\ \alpha \end{bmatrix}$, where $\alpha =$

(a) 2

(b) -3

(c) 1

(d) 4

(e) -1

13. The eigenvector associated with the eigenvalue $\lambda = 4$ of the matrix $A = \begin{bmatrix} 8 & -5 \\ 4 & -1 \end{bmatrix}$ is $\begin{bmatrix} a \\ 4 \end{bmatrix}$, where $a =$

(a) 6

(b) 7

(c) -5

(d) 5

(e) -4

14. Consider the subspace S of $\mathbb{R}^4$ defined by $S = \{(a, b, c, d) \mid a + 8c = b + 7d = 0\}$. A basis of S consists of the vectors

(a) $v_1 = (-8, 0, 1, 0),\ v_2 = (0, 7, 0, 0)$

(b) $v_1 = (-8, 0, 1, 0),\ v_2 = (0, -7, 0, 1)$

(c) $v_1 = (8, 0, 1, 0),\ v_2 = (0, -7, 0, 1)$

(d) $v_1 = (-8, 0, 2, 0),\ v_2 = (0, 1, 0, -7)$

(e) $v_1 = (-8, 1, 0, 1),\ v_2 = (-7, 0, 1, 0)$

15. The general solution of the differential equation $4y'' + 4y' + y = 0$ is

(a) $y(x) = c_1 e^{-x} + c_2 e^{-x/2}$

(b) $y(x) = (c_1 + c_2 x)e^{2x}$

(c) $y(x) = c_1 + c_2 x e^{-x/2}$

(d) $y(x) = (c_1 + c_2 x)e^{-x/2}$

(e) $y(x) = (c_1 + c_2 x)e^{-x}$

16. All values of k for which the following vectors

$$v_1 = (7, 1, -8),\ v_2 = (-4, 2, -7) \text{ and } v_3 = (3, k, 3)$$

form a basis for $\mathbb{R}^3$ are

(a) $k \neq -4$

(b) $k \neq -2$

(c) $k \neq 3$

(d) $k \neq -1$

(e) $k \neq 2$

17. If $y_p = Ae^x + Bxe^x$ is a particular solution of the differential equation

$$4y'' + 4y' + y = 3xe^x,$$

then $B =$

(a) $\frac{1}{2}$

(b) 3

(c) 2

(d) 4

(e) $\frac{1}{3}$

18. If the characteristic polynomial of the matrix $A = \begin{bmatrix} 3 & 6 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is $P(\lambda) = -(\lambda - 1)^2(\lambda - 3)$, then a basis for the eigenspace of $\lambda = 1$ is $v_1 = \begin{bmatrix} 1 \\ 0 \\ \alpha \end{bmatrix}$ and $v_2 = \begin{bmatrix} \beta \\ 1 \\ 0 \end{bmatrix}$, then $\alpha + \beta =$

(a) 0

(b) 2

(c) -2

(d) 4

(e) -4

19. Using the method of variation of parameters, a particular solution of the differential equation $y'' + 9y = \sin(3x)$ is $y_p(x) =$

(a) $\frac{1}{8}x\sin(3x)$

(b) $\frac{1}{36}x\cos(3x)$

(c) $-\frac{1}{6}x\cos(3x)$

(d) $-\frac{1}{6}x\sin(3x)$

(e) $\frac{1}{36}x\sin(3x)$

20. The general solution of the differential equation $y^{(4)} = 16y$ is

(a) $y(x) = (c_1 + c_2x)e^{2x} + c_3\cos(2x) + c_4\sin(2x)$

(b) $y(x) = c_1e^{2x} + c_2e^{-2x} + c_3x\cos(2x) + c_4x\sin(2x)$

(c) $y(x) = (c_1 + c_2x)e^{-2x} + c_3\cos(2x) + c_4\sin(2x)$

(d) $y(x) = c_1e^{2x} + c_2e^{-2x} + c_3\cos(2x) + c_4\sin(2x)$

(e) $y(x) = c_1e^{2x} + c_2e^{x} + c_3\cos(2x) + c_4\sin(2x)$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(a)**	$r = 2$ is a simple root, so Axe^{2x}; $r = 0$ is not a root, so $Bx + C$.
2	**(d)**	Row reduction leaves 3 nonzero rows, so $\operatorname{rank}(A) = 3$.
3	**(d)**	$r^4 + r^3 - 3r^2 - 5r - 2 = (r-2)(r+1)^3$: $A = 2,\ B = -1$, so $A - B = 3$.
4	**(a)**	$W = e^{2x}\left[-1 + 4(\cosh^2 x - \sinh^2 x)\right] = 3e^{2x}$.
5	**(c)**	$p = (5-\lambda)\lambda(\lambda - 2) = -\lambda^3 + 7\lambda^2 - 10\lambda$.
6	**(a)**	$(r-3)^2(r^2+16) = r^4 - 6r^3 + 25r^2 - 96r + 144$.
7	**(b)**	u in the system gives $\alpha = 1$, v gives $\beta = -5$, so $\alpha - \beta = 6$.
8	**(e)**	$\sinh x = \frac{1}{2}(e^x - e^{-x})$, so the first set is dependent.
9	**(e)**	$r = \pm 2i$; $c_1 = 3$, $2c_2 = 8$: $y = 3\cos 2x + 4\sin 2x$.
10	**(d)**	$-4a + 3b = 7,\ 2a - b = 1$ give $a = 5,\ b = 9$, so $a + b = 14$.
11	**(b)**	$\lambda = 1, 2$ with eigenvectors $(1,1), (4,3)$: $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \operatorname{diag}(1,2)$.
12	**(c)**	$iv_1 + v_2 = 0$ with $v_1 = i$ gives $v_2 = -i \cdot i = 1$, so $\alpha = 1$.
13	**(d)**	$(A - 4I)v = 0$: $4v_1 = 5v_2$, so $v = (5,4)$ and $a = 5$.
14	**(b)**	$a = -8c,\ b = -7d$: basis $(-8,0,1,0),\ (0,-7,0,1)$.
15	**(d)**	$4r^2 + 4r + 1 = (2r+1)^2$, repeated root $-\frac{1}{2}$: $y = (c_1 + c_2 x)e^{-x/2}$.
16	**(d)**	$\det = 81(k+1) \neq 0$, so $k \neq -1$.
17	**(e)**	Substituting: $9B = 3$ and $9A + 12B = 0$, so $B = \frac{1}{3}$.
18	**(c)**	$A - I$ gives $x + 3y - z = 0$: $\alpha = 1,\ \beta = -3$, so $\alpha + \beta = -2$.
19	**(c)**	$W = 3$; $u_1 = -\frac{x}{6} + \ldots$, so $y_p = -\frac{1}{6}x\cos 3x$ (dropping a homogeneous term).
20	**(d)**	$r^4 - 16 = (r^2-4)(r^2+4)$: roots $\pm 2, \pm 2i$.